

22 63 00 - Gas Systems for Laboratory and Healthcare Facilities

For Laboratory Buildings with a B Occupancy classification, verify that the quantity of stored gases classified as hazardous materials does not exceed the exempt quantities called out in the Building Code. H occupancy classification may otherwise be required.

Systems and storage for toxic gases shall be in compliance with the Uniform Fire Code including California Amendments in particular Section 8003, covering toxic compressed gases. Additionally, such systems shall be in compliance with the Toxic Gas Ordinance as written by the Western Fire Chiefs Association. Inform both the Executive Architect and University's Representative of requirements for gas storage cabinets, double containment piping, monitoring systems, and maximum quantity limitations in order to assure coordination among all disciplines.

Storage and piping concepts for laboratory gasses should be reviewed with Cal Poly Environmental Health and Safety via the University's Representative early in the design process. Issues include the possibility of asphyxiation (includes inert gasses) in confined spaces, as well as hazardous material concerns.

Laboratory gasses will not be used or stored in rooms having a HVAC system which returns air for recirculation.

Provide seismic restraint systems for all gas canisters and dewars.

Acceptable Piping Materials

Inert, Nontoxic Gases (nitrogen): Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing shall be pre-washed, degreased and capped at both ends by the manufacturer. Fittings shall be pre-washed, degreased and wrapped by the manufacturer. All joints shall be brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to prevent oxidation.

Other Gasses: Piping materials shall be selected on a project by project basis to meet specific program, compatibility, and regulatory requirements. Design for all toxic or hazardous gasses shall be reviewed and approved by EH&S.

Valving:

Laboratory gas systems shall be provided with manual isolation valves at the following locations:

At each floor where branch piping connects to a riser.

At the gas canister connections.

At branch connections to mains.

Upstream of any equipment connected to the laboratory gas system.